

1. The table shows the date and exchange rate for the Japanese Yen.

Date	Exchange Rate
October 8, 1982	267.38 Yen = 1 US dollar
June 30, 1995	84.90 Yen = 1 US dollar

- a. Determine the cost, in US dollars, for a souvenir that cost 2,500 Yen on October 8, 1982. Show your work in the provided space.

Souvenir Cost October 8, 1982 (US dollars)	Work
\$9.35	$2500 \text{ Yen} \cdot \frac{\$1}{267.38 \text{ Yen}}$

- b. Determine the cost, in Yen, for a hotel room that cost \$99 on June 30, 1995. Show your work in the provided space.

Hotel Room Cost June 30, 1995 (Yen)	Work
8405.10	$\$99 \cdot \frac{84.90 \text{ Yen}}{\$1}$

2. Use conversion rates to find the value. Show your work in the provided space. Round to the nearest hundredth, if necessary.

7 ounces = 198.45 OR 198.86 grams
Work
$7 \text{ oz} \cdot \frac{28.35 \text{ g}}{1 \text{ oz}}$ OR $7 \text{ oz} \cdot \frac{1 \text{ g}}{0.0352 \text{ oz}}$

3. A hallway has a length of 350 inches (in). and width of 90 in. What are the dimensions of the hallway in meters (m)? Show your work in the provided space. Round to the nearest thousandth, if necessary.

Length (m)	Width (m)
8.89 OR 8.974	2.286 OR 2.308
Work	Work
$350 \text{ in.} \cdot \frac{2.54 \text{ cm}}{1 \text{ in.}} \cdot \frac{1 \text{ m}}{100 \text{ cm}}$ OR $350 \text{ in.} \cdot \frac{1 \text{ cm}}{0.39 \text{ in.}} \cdot \frac{1 \text{ m}}{100 \text{ cm}}$	$90 \text{ in.} \cdot \frac{2.54 \text{ cm}}{1 \text{ in.}} \cdot \frac{1 \text{ m}}{100 \text{ cm}}$ OR $90 \text{ in.} \cdot \frac{1 \text{ cm}}{0.39 \text{ in.}} \cdot \frac{1 \text{ m}}{100 \text{ cm}}$

4. Use conversion rates to find the value. Round to the nearest thousandth, if necessary.

$$5 \text{ cups} = 1200 \text{ OR } 1182.033 \text{ milliliters}$$

Work

$$5 \text{ cups} \cdot \frac{0.24 \text{ liters}}{1 \text{ cup}} \cdot \frac{1000 \text{ mL}}{1 \text{ liter}}$$

OR

$$5 \text{ cups} \cdot \frac{1 \text{ liter}}{4.23 \text{ cups}} \cdot \frac{1000 \text{ mL}}{1 \text{ liter}}$$

5. All of a person's blood vessels can stretch out to approximately 100,000 kilometers.

Determine the length of a person's blood vessels in miles.

Show your work in the provided space. Round to the nearest tenth of a mile, if necessary.

Length

$$62111.8 \text{ OR } 62000 \text{ miles}$$

Work

$$100000 \text{ km} \cdot \frac{1 \text{ mile}}{1.61 \text{ km}}$$

OR

$$100000 \text{ km} \cdot \frac{0.62 \text{ mile}}{1 \text{ km}}$$

6. Hummingbird feeders are filled with hummingbird nectar. Hummingbird nectar is made using a ratio of 1 part sugar to 4 parts water.

How much sugar, in ounces, is needed to make 27 ounces of hummingbird nectar? Show your work in the provided space.

Sugar
5.4 ounces
Work
$\frac{1}{5} = \frac{\text{sugar}}{\text{total}}$ $\frac{1}{5} = \frac{x \text{ ounces of sugar}}{27 \text{ ounces}}$

7. In 2010 a black bear named M34 was tracked through forests, neighborhoods, and state parks across Central Florida logging approximately 500 miles in 56 days. For 25.3 miles of M34's journey, he was tracked walking along the Kissimmee River.

How many days was M34 traveling along Kissimmee River?

Show your work in the provided space. Round to the nearest tenth of a day.

Kissimmee River	Work
2.8 days	$\frac{56 \text{ days}}{500 \text{ miles}} = \frac{x \text{ days traveling along the river}}{25.3 \text{ miles}}$

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8. A recipe for 120 ounces (oz) of chocolate milk is shown.
- 120 oz of cold milk
 - 20 teaspoons (tsp) Dutch-processed cocoa powder
 - 21 tsp of sugar

How many teaspoons of cocoa powder are needed to make 16 oz of chocolate milk? Show your work in the provided space.

Cocoa Powder	Work
$2\frac{2}{3}$ tsp	$\frac{20 \text{ tsp}}{120 \text{ oz}} = \frac{x \text{ tsp of cocoa powder}}{16 \text{ oz}}$

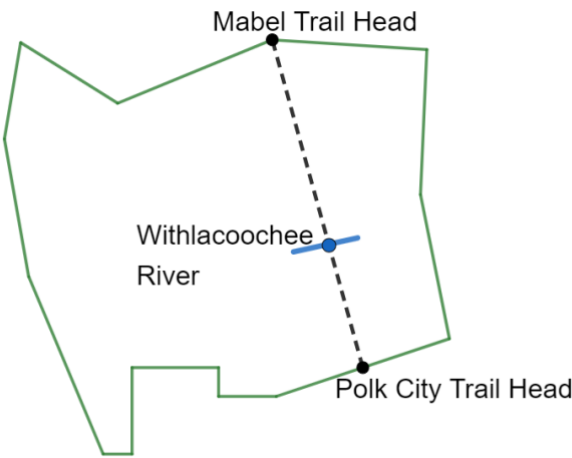
9. When the height of the sand in a particular rectangular sandbox is leveled out then the height of the sand, in inches (in.), is proportional to the volume of sand, in cubic inches (in.³), in the sandbox. When the height of the sand is 2.25 in. the volume of the sand is 180 in.³. A playground has 3 of these sandboxes.

What is the total volume of the sand, in cubic inches (in.³), that is needed for the playground when the height of the sand in each sandbox is 3.5 in.?

Show your work in the provided space.

Volume	Work
840 in.^3	$\frac{2.25 \text{ in}}{180 \text{ in}^3} = \frac{3.5 \text{ in}}{x \text{ in}^3}$ $x = 280 \text{ in}^3$ $280 \text{ in}^3 \times 3 = 840 \text{ in}^3$

10. Four important Florida rivers originate in the Green Swamp – the Peace, the Withlacoochee, the Ocklawaha, and the Hillsborough. The Van Fleet bike trail is a 29.2 mile bike trail that goes through the Green Swamp. A map that shows the location of the bike trail’s two trail heads and the bridge that crosses the Withlacoochee River is shown. The table gives the measurement, in centimeters (cm), for the locations on the map.

Map	Measurement
	Polk City Trail Head to Withlacoochee River 4.4 cm
	Polk City Trail Head to Mabel Trail Head 11.8 cm

What is the distance, in miles, from the Polk City Trail Head to the Withlacoochee River?

Show your work in the provided space. Round to the nearest tenth of a mile.

Distance	Work
10.9 miles	$\frac{4.4 \text{ cm}}{11.8 \text{ cm}} = \frac{x \text{ miles from Polk City Trail Head to the Withlacoochee River}}{29.2 \text{ miles}}$